TrueML Data Science Exercise
The data included in this exercise is related to propensity to make a payment on a debt.

Please submit your analysis and findings, preferably in notebook format, and be prepared to walk through the exercise in your interview.

Fields include:

column	description
account_id	ID of the account
made_last_payment	Whether account made their latest payment on time (outcome)
latest_communication_channel	The channel (e.g., email/SMS) used for the most recent communication to the account
minimum_payment	The minimum payment amount due on the account
previous_payment_amount	The dollar amount of the account's most recent payment
product	The product type associated with the account (Credit Card or Deferred Repayment)
last_reminder_sent_days	Number of days since the last payment reminder was sent to the account
debt_to_income_ratio	Estimated ratio of total debt to income
opened_last_communication	Whether the account holder opened/read the most recent communication sent to them
used_chat_feature	Whether the account holder has used the assistance chat feature on the website
count_comms_sent_last_30d	Total number of communications sent to the account in the last 30 days
homeowner	Whether the account holder is a homeowner
ever_missed_payment	Whether the account has any prior missed payment on record
age_of_debt_yrs	Age of the debt in years since origination
total_balance	Total outstanding balance on the account in dollars
latest_communication_dow	Day of week of latest communication (Mon-Fri)

Questions to address:

- Explore and summarize the data using descriptive statistics and/or plots (this doesn't need to be comprehensive, pick a few things to explore).
    - Explain what you're seeing and how it informs your understanding of the data. Do you observe any noteworthy patterns or insights?
- Write a function or set or functions to run comparisons across product type. The data includes only two products, but assume other products exist and you may want to do comparisons between them. Include a unit test for your function.
- Build a model to predict a customer's probability of paying their next bill (`made_last_payment`).
    - We are most interested in the steps you would take to build a good predictive model—not that you have built the best possible model.

We are interested in learning how **you** approach problems and data, so please refrain from using AI for this exercise.

NOTE: This is simulated data (so don't worry if you find results that seem unintuitive or "off")